



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

This is how we organize data!



Learning Outcome:

Exchange written and spoken information on principles, concepts, data base objects related to relational database management system in spoken or written texts.
Describe relational database, its concepts, and principles in spoken or written texts.

Ask your partner:

- Why do you think it is important to learn about the different data models that exist?
- What criteria do people follow to choose one data model over another? Share your thoughts.

Can Do Statements:

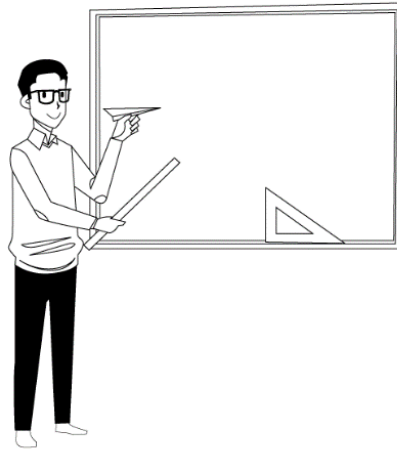
- I can say what data models do and what they are.
- I can talk about the entity relationship model, ER.
- I can discuss briefly the relational data model.





Let's Explore!

Getting Prepared for the Exam



Instructions: Read the text below.

- Mr. Goodman:** Good morning! Before presenting the exam, we can make a short review of the topics we checked this week. Data models, for example, might be included in the exam.
- Brian:** Can we discuss what they are and do, please?
- Mr. Goodman:** Sure. Data models are really important for DBMS. These models basically shape the logical structures of the systems as well as provide abstraction qualities to the databases they work on.
- Brian:** I think a simpler explanation may be required...
- Mr. Goodman:** Okay. This model might be perceived as the pillar of the database as it shapes the way the information is linked within databases. Furthermore, it also plays a big role in how the system processes and stores data.
- Brian:** That's pretty clear now. Do you agree that by not covering ER models, this topic is incomplete?
- Mr. Goodman:** That is true. Now, making connections among world entities and attributes might be seen as one of the main advantages of using the entity relationship model. In fact, by not leaving aside two important concepts, we can get to understand this topic. Those concepts are entities, units with their own attributes called domains, and relationships, which are expressed through graphs showing mapping cardinalities.
- Brian:** Great! What about the relational data model?
- Mr. Goodman:** Well, that topic might be also included in the exam.
- Brian:** I remember that this model may be identified as one of the simplest and clearest models to use as it uses relational tables.
- Mr. Goodman:** That is correct. This model allows us to create and analyze inter-related tables; in other words, we can unequivocally check entities as well as their attributes. In fact, this model is the most scientific and clear out there.





Let's Practice!

Early data models may be seen as not so scientific...

Instructions: Match the phrases from column A to column B to complete the sentence.

COLUMN A

1. Data models might be seen
2. Logical structures may be
3. Abstraction qualities may
4. Regarding information, the chosen data model might be seen as
5. Finally, the pillar of any system might be perceived

COLUMN B

- a. to be the data model as they are in charge of the processing and storing of data.
- b. be defined depending on the data model.
- c. shaped by the data model that is implemented.
- d. as important elements for any DBMS.
- e. the responsible of the way the system links values and data.



Let's Practice!

'ER' makes reference to the entity relationship model.

Instructions: Select the correct answer from the box in order to complete the sentence.

world entities and attributes / attributes or domains / entities and relationships / ER models / relationships and mapping cardinalities / graphs and charts / simple and clear

1. By not covering _____, the topic of data models is definitely incomplete.
2. By not leaving aside two important terms, _____, ER models are easily understood.
3. By not forgetting that entities are basically values that have _____, we can get to understand what the entity relationship model does.
4. By not ignoring that _____ are strongly related, we can understand ER models better.
5. Finally, by not leaving aside the fact that ER models make connections among _____, we can get to comprehend how these models work.



Extension Activity

Investigation More About Data Models...

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about data models.
- Create a poster with the main characteristics you found about data models.
- Share and discuss with your partners the poster you created.

Self-Assessment

Yes

No

- I look for more information about data models. ☐ Yes ☐ No
- I create a poster with the main characteristics of data models. ☐ Yes ☐ No
- I share the poster I created with my partners. ☐ Yes ☐ No



Target Words

- **mapping cardinality** (noun): factor
- **abstraction** (noun): a process in which irrelevant values or pieces of information are not displayed in a database, making it easier for the user to navigate it
e.g. These models basically shape the logical structures of the systems as well as provide abstraction qualities to the databases they work on.
- **pillar** (noun): a highly important element; the base of a concept, term or process
e.g. This model might be perceived as the pillar of the database as it shapes the way the information is linked within databases.
- **linked** (adjective): connected, joined to another element
e.g. This model might be perceived as the pillar of the database as it shapes the way the information is linked within databases.

- **mapping cardinality** (noun): factor that delimits the specific amount of associations that an entity can have with another
e.g. Those concepts are entities, units with their own attributes called domains, and relationships, which are expressed through graphs showing mapping cardinalities.
- **inter-related** (adjective): connected, linked to other elements
e.g. This model allows us to create and analyze inter-related tables; in other words, we can unequivocally check entities as well as their attributes.
- **unequivocally** (adverb): something that is clear and without any contradiction
e.g. This model allows us to create and analyze inter-related tables; in other words, we can unequivocally check entities as well as their attributes.



Learning Points

Passive of Modals May and Might

We use the passive form to **highlight the element that receives the action in a sentence** instead of its subject (the doer of the action). Now, don't forget that **modal verbs can also have their own passive form**.

You use modals in the passive voice to change the intention of the sentence. Remember that both '**may**' as well as '**might**' express possibility in the future. In fact, '**may**' tends to imply the idea that something is more likely to occur than '**might**'. Furthermore, '**might**' tends to be perceived as the most formal one.

The structure that we follow to make sentences using the passive voice of the modal verbs 'may' and 'might' is:

Object of the sentence + may / might + be + action verb (in past participle) + (by + subject*) + complement.

*Remember that in the passive voice, you can omit the subject of the sentence; however, when you want to include it, you just have to mention it in this part of the sentence.

For example:

*A data model **might be used** to organize all this information.*

*A meeting **may be called** to discuss data models.*

*The DBMS model **might be developed by the designers**.*

Examples

- *I think a simpler explanation **may be required**...*
- *Well, that topic **might be also included in the exam**.*



Learning Points

Relational Tables

The relational database model is mainly made using a table to present all the information.

The relational table is successful because as it is organized logically, it is easier to understand; in fact, this model is logic-driven, which makes it more intuitive and natural. This type of model is composed by relations, or tables, that present rows (attributes) and columns. This distribution allows the data contained in them to be related with one another.

In the table, the rows represent a single entity (for example, a person), while the columns represent an attribute of that single entity. Then, each crossing represents a particular data value.

Check the following chart as it presents an example of a relational table for data in a school:

Stu_ID	Stu_Lname	Stu_Fname	Stu_GPA
22275	Gobble	Anne	32
22276	Gomez	Carlos	45
22277	Gorth	Lauren	22